

NF- κ B Is Required for UV-Induced JNK Activation via Induction of PKC δ

Jing Liu,¹ Dan Yang,¹ Yuzuru Minemoto,¹
Michael Leitges,² Marsha R. Rosner,¹
and Anning Lin^{1,*}

¹Ben May Institute for Cancer Research
The University of Chicago
929 East 57th Street
Chicago, Illinois 60637

²Max Planck Institute for Experimental Endocrinology
Hanover
Germany

Summary

Ultraviolet (UV) exerts its biological activities by activating downstream effectors, including NF- κ B, JNK, and caspases. Activation of JNK is required for UV-induced apoptosis. It is unknown whether any crosstalk occurs between NF- κ B and JNK in response to UV and, if so, how it affects UV killing. Here we report that NF- κ B promotes UV-induced JNK activation, thereby contributing to UV-induced apoptosis. UV-induced JNK activation is impaired in RelA/NF- κ B null murine embryonic fibroblasts. In resting cells, the preexisting nuclear RelA has already been recruited to PKC δ promoter and is essential for its expression. UV-induced rapid and robust activation of JNK requires PKC δ , which augments JNK phosphorylation-activation by its upstream kinases. The RelA/NF- κ B-PKC δ -JNK pathway is critical for UV-induced apoptosis, as it induces the immediate expression of the proapoptotic Fas ligand. Thus, our results demonstrate that RelA/NF- κ B via PKC δ positively regulates UV-induced JNK activation and provide a mechanism by which NF- κ B promotes UV-induced apoptosis.

Introduction

c-Jun N-terminal protein kinase (JNK; also known as SAPK) (Hibi et al., 1993) is a member of the mitogen-activated protein kinase (MAPK) superfamily. JNK has two ubiquitously expressed isoforms, JNK1 and JNK2, and a tissue-specific isoform JNK3, with different splicing forms (p54 and p46) (Chang and Karin, 2001; Davis, 2000; Lin, 2003). JNK is activated by sequential protein phosphorylation through a MAP kinase module, i.e., MAPK kinase kinase (MAP3K) $\rightarrow$ MAPK kinase (MAP2K or MKK) $\rightarrow$ MAPK, in response to a variety of extracellular stimuli (Karin, 1995; Lin, 2003). Two MAP2Ks, JNK-activating kinase 1 and 2 (JNKK1/MKK4 and JNKK2/MKK7), and several MAP3Ks are involved in JNK activation (Lin, 2003). Recent studies show that JNK1 but not JNK2 is significantly activated by extracellular stimuli including the proinflammatory cytokine tumor necrosis factor (TNF- α) and the physical stress ultraviolet (UV) (Liu et al., 2004; Sabpathy et al., 2004). Once activated, JNK phosphorylates and regulates the activity of several transcription factors, such as c-Jun,

ATF-2, Elk-1, p53, and c-Myc as well as nontranscription factors such as members of the Bcl-2 family proteins (Lin, 2003; Yu et al., 2004). Overwhelming evidence shows that JNK has a central role in regulating many cellular activities from growth control to apoptosis (Chang and Karin, 2001; Davis, 2000; Lin, 2003).

The dimeric NF- κ B transcription factors are key regulators in immune responses, inflammation, infections, oncogenesis, and apoptosis (Baldwin, 1996; Ghosh et al., 1998; Karin and Lin, 2002; Lenardo and Baltimore, 1989; Lin and Karin, 2003; Thanos and Maniatis, 1995; Verma et al., 1995). Among five mammalian NF- κ B proteins, RelA/p65, c-Rel, RelB, NF- κ B1 (p50/p105), and NF- κ B2 (p52/p100), RelA is the major transactivating subunit of NF- κ B (Baldwin, 1996). In nonstimulated cells, the majority of NF- κ B is retained in the cytoplasm via interaction with its inhibitors (I κ Bs) (Baldwin, 1996). Yet a small amount of NF- κ B proteins including RelA reside in the nucleus as the result of constitutive degradation of I κ B proteins (Pando and Verma, 2000). This preexisting nuclear NF- κ B activity may be important for the expression of certain NF- κ B target genes that are involved in maintaining normal cellular functions and priming cells to extracellular stimuli. Upon stimulation by most NF- κ B-inducing stimuli such as TNF- α , I κ Bs are rapidly phosphorylated by the I κ B kinase (IKK) complex, followed by ubiquitination-dependent degradation by the 26S proteasome (Karin and Ben-Neriah, 2000; Ghosh and Karin, 2002). One of the exceptions is UV, which induces slow proteasome-dependent degradation of I κ B α (Bender et al., 1998; Li and Karin, 1998) via the p38-CK2 pathway (Kato et al., 2003). Freed NF- κ B translocates into the nucleus, where it further stimulates expression of target genes involved in many important cellular events (Baldwin, 1996; Ghosh et al., 1998; Lenardo and Baltimore, 1989; Thanos and Maniatis, 1995; Verma et al., 1995).

NF- κ B regulates the activities of many signaling pathways in the intracellular signaling network, thereby playing a critical role in determining cellular responses to extracellular stimuli (Karin and Lin, 2002; Lin and Karin, 2003; Lin, 2003). One such crosstalk is the inhibition of caspase and JNK by NF- κ B in response to TNF- α . TNF- α induces activation of NF- κ B, JNK, and caspases (Karin and Lin, 2002; Lin, 2003). However, NF- κ B-induced inhibitors of apoptosis (IAPs) inhibit caspases, thereby suppressing TNF- α -induced apoptosis (Beg and Baltimore, 1996; Liu et al., 1996; Van Antwerp et al., 1996; Wang et al., 1996). NF- κ B also prevents TNF- α -induced JNK1 activation, which otherwise contributes to TNF- α killing (Kamata et al., 2005; Karin and Lin, 2002; Lin, 2003; Liu et al., 2004; Maeda et al., 2003; Tang et al., 2001a; Tang et al., 2002). However, the constitutively active JNK itself is insufficient to induce apoptosis (Tang et al., 2001a; Tang et al., 2002; Lin, 2003). Furthermore, JNK suppresses apoptosis induced by withdrawal of the hematopoietic cytokine interleukin-3 (IL-3) via phosphorylation and inactivation of the proapoptotic Bcl-2 family protein BAD (Yu et al., 2004). Thus, JNK has pro- and antiapoptotic functions,

*Correspondence: alin@huggins.bsd.uchicago.edu

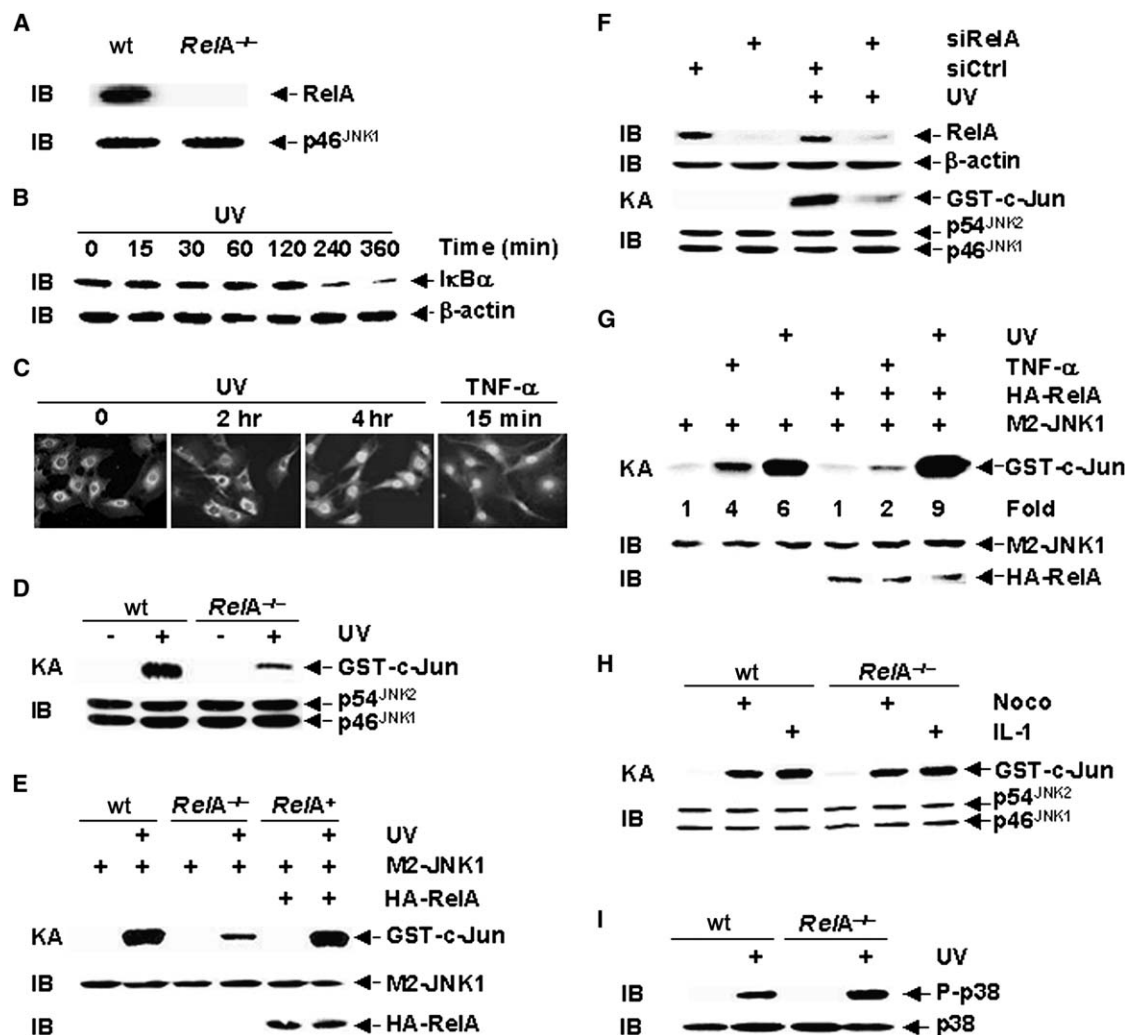


Figure 1. The Preexisting Nuclear RelA Is Required for UV-Induced JNK Activation

(A) Immunoblotting (IB) of RelA and JNK in wt and *RelA*^{-/-} mouse fibroblasts.

(B and C) Wt fibroblasts were irradiated with UV (20 J/m²) for various times or TNF- α (5 ng/ml, 15 min), as indicated. IkB α degradation was analyzed by immunoblotting (B), and RelA nuclear translocation was determined by indirect immunofluorescence analysis (C).

(D) Wt and *RelA*^{-/-} cells were irradiated with or without UV (20 J/m², 30 min). Activation and expression of JNK were measured by immune complex kinase assays (KA) and immunoblotting, respectively.

(E) Wt or *RelA*^{-/-} cells were transfected with expression vectors encoding M2-JNK1 (1 μ g) along with HA-RelA or empty vector (0.1 μ g each). After 40 hr, cells were irradiated with or without UV (20 J/m², 30 min). M2-JNK1 activity and expression of M2-JNK1 and HA-RelA were determined.

(F) Wt cells were transfected with the control or RelA siRNA (200 nM each). UV-induced JNK activation and expression of JNK and RelA were determined as in (A) and (D).

(G) Wt fibroblasts were transfected with expression vectors encoding M2-JNK1 (1 μ g) along with HA-RelA or empty vector (0.1 μ g). After 40 hr, cells were stimulated with or without TNF- α (5 ng/ml, 15 min) or UV (20 J/m², 30 min). M2-JNK1 activity and expression of M2-JNK1 and HA-RelA were determined.

(H and I) Wt and *RelA*^{-/-} cells were treated with or without nocodazole (0.5 μ g/ml, 30 min) or IL-1 β (2 ng/ml, 15 min) (H) or UV (20 J/m², 30 min) (I). Activation and expression of JNK or p38 were determined.

depending on the death stimulus and the cell context (Lin, 2003). Like TNF- α , UV activates NF- κ B, JNK, and caspases (Herrlich et al., 1997). However, UV induces rapid and robust activation of JNK, which is essential for UV killing (Liu et al., 2004; Tournier et al., 2000), but slow activation of NF- κ B (Bender et al., 1998; Li and Karin, 1998). It is unknown whether NF- κ B regulates UV-induced JNK activation and, if so, how it affects UV-induced apoptosis. Here we report that the preexisting nuclear RelA/NF- κ B positively regulates UV-induced JNK activation via induction of protein kinase C δ (PKC δ).

This positive crosstalk, which is specific to the UV signaling, contributes to UV-induced apoptosis.

Results

The Preexisting Nuclear RelA Is Required for UV-Induced JNK Activation

To test whether any crosstalk occurs between NF- κ B and JNK in response to UV irradiation, we used wild-type (wt) and RelA-deficient (*RelA*^{-/-}) mouse embryonic fibroblasts (Figure 1A). In wt fibroblasts, NF- κ B

activation by UV was slow, taking more than 2 hr, as measured by proteasome-mediated I κ B α degradation (Figure 1B) and RelA nuclear translocation (Figure 1C). By contrast, UV-induced JNK activation was rapid and robust, reaching its maximum 30 min postirradiation, as determined by immune complex kinase assays (Figure 1D; data not shown), consistent with previous reports (Bender et al., 1998; Li and Karin, 1998). Interestingly, UV-induced JNK activation was significantly reduced in *RelA*^{-/-} cells (Figure 1D, 17% of that in wt fibroblasts), and so was UV-induced c-Jun transcription activity (data not shown). Introduction of HA-RelA into *RelA*^{-/-} cells restored UV-induced M2-JNK1 activation (Figure 1E), suggesting that the reduced JNK activation was due to loss of RelA. Transfection of wt fibroblasts with RelA siRNA, which inhibited RelA expression (Figure 1F) but not the control scramble siRNA, also suppressed UV-induced JNK activation (Figure 1F). Conversely, transfection of HA-RelA into wt fibroblasts further promoted cotransfected M2-JNK1 activation by UV (Figure 1G). Under the same conditions, TNF- α -induced activation of M2-JNK1 was inhibited (Figure 1G), as previously reported (Tang et al., 2001a). Loss of RelA only affected JNK activation by UV but not IL-1 β or nocodazole (Figure 1H) or UV-induced p38 activation (Figure 1I). These data demonstrated that the preexisting nuclear RelA (RelA hereinafter unless indicated otherwise) was required for rapid and robust JNK activation by UV.

Expression of PKC δ Depends on RelA

It is possible that RelA controls the expression of certain key component(s) in the UV-JNK signaling pathway, thereby promoting UV-induced JNK activation. To test this scenario, we analyzed several candidates, including epithelial growth factor receptor (EGF-R) (Rosette and Karin, 1996), TNF-receptor II (TNF-RII) (Rosette and Karin, 1996), TNF-receptor-associated factor 2 (TRAF2) (Shi and Kehrl, 1997), and various PKC isoforms (Kawakami et al., 1998). Immunoblotting analysis revealed that, among the above candidates, only PKC δ protein levels were severely reduced in *RelA*^{-/-} cells or in RelA siRNA-expressing wt fibroblasts under nonstimulated conditions (Figures 2A and 2B; data not shown). Introduction of HA-RelA into *RelA*^{-/-} cells partially restored PKC δ expression (Figure 2A), suggesting that the reduction of PKC δ expression in *RelA*^{-/-} cells was due to loss of RelA. Consequently, PKC δ activation by UV in *RelA*^{-/-} cells was impaired, as measured by immune complex kinase assays (Figure 2C). The reduced PKC δ protein levels were due to its impaired transcription, as measured by semiquantitative RT-PCR (Figure 2D). Quantitative real-time RT-PCR revealed that PKC δ mRNA levels in *RelA*^{-/-} cells were only one third of those in wt fibroblasts (Figure 2E). Reintroduction of RelA into *RelA*^{-/-} cells partially restored expression levels of PKC δ mRNA (Figures 2D and 2E) but failed to do so in the presence of RelA siRNA (Figure 2D). Taken together, RelA/NF- κ B is essential for PKC δ transcription and protein expression.

To determine whether PKC δ is an NF- κ B target gene, we analyzed PKC δ promoter. Two putative NF- κ B binding sites (PKC δ - κ B1, at position -83 to -74, and PKC δ - κ B2, -52 to -43 of PKC δ gene) were identified using

TFSEARCH program (Version 1.3). Electrophoresis mobility shift assays (EMSA) revealed that ³²P-labeled oligonucleotide probes comprising either PKC δ - κ B1 or - κ B2 site formed complexes with proteins prepared from nonstimulated wt fibroblasts, and the DNA binding activity was further enhanced by TNF- α (Figure 2F) or UV (data not shown). The protein-DNA complexes were specific, as they were competed away by unlabeled probes (Figure 2F). Under the same conditions, no protein-DNA complexes were detected with the probes containing mutations at the NF- κ B sites (data not shown). The protein-DNA complexes were retarded by anti-RelA antibody (Figure 2F), suggesting that they contained RelA. In nonstimulated wt fibroblasts, endogenous RelA had already been recruited to PKC δ promoter, as analyzed by chromatin immunoprecipitation (ChIP) assays (Figure 2G). The recruitment was further enhanced by UV and TNF- α (Figure 2G). Fusion of the PKC δ promoter (-309 to -1 that contains PKC δ - κ B1 and - κ B2 sites) upstream to a promoterless luciferase reporter gene efficiently drove the expression of the latter in nonstimulated cells, and the promoter activity was further stimulated by cotransfected HA-RelA (Figure 2H) or by IL-1 β in an NF- κ B-dependent manner (Figure 2I), as measured by the LUC-reporter gene assays. Interestingly, mutation of PKC δ - κ B1 site in the PKC δ promoter had no detectable effect on basal but abolished the stimulated LUC-reporter gene activity (Figures 2H and 2I). By contrast, mutation of PKC δ - κ B2 site abolished both basal and stimulated activities of the LUC-reporter gene (Figures 2H and 2I). Thus, the κ B1 site was likely involved in the stimulated PKC δ promoter activity, whereas the κ B2 site was likely involved in both basal and stimulated activity of the PKC δ promoter. Taken together, PKC δ is a bona fide RelA/NF- κ B target gene.

To understand how PKC δ expression is regulated by RelA/NF- κ B, we examined expression profiles of PKC δ and several other NF- κ B target genes (Cox-2, I κ B α , and TRAF2) in nonstimulated wt and *RelA*^{-/-} fibroblasts. Immunoblotting analysis revealed that protein levels of PKC δ and Cox-2 were already severely reduced in *RelA*^{-/-} cells (Figure 3A). By contrast, protein levels of I κ B α and TRAF2 were not affected by loss of RelA (Figure 3A). mRNA levels of PKC δ and Cox-2 but not I κ B α and TRAF2 were also significantly reduced in *RelA*^{-/-} cells, as measured by semiquantitative RT-PCR and quantitative real-time RT-PCR (Figures 3B and 3C). ChIP assays revealed that, in nonstimulated wt fibroblasts, RelA had already been recruited to the promoters of PKC δ and Cox-2, and the recruitment was stimulated by UV and TNF- α (Figure 3D). Under the same conditions, the recruitment of RelA to the promoters of I κ B α and TRAF2 was negligible, although it was stimulated by UV and TNF- α (Figure 3D). The difference in the recruitment of RelA to the above promoters was not due to variations in the chromatin amount used (Figure 3D), nor was it the result of the difference in PCR efficiency, as TNF- α -induced recruitment of RelA to all four promoters above was similar (Figure 3D). Like RelA, RelB had also been recruited to the promoter of PKC δ , but not I κ B α , under nonstimulated conditions (Figure 3E), and the recruitment of RelB to both promoters was enhanced by UV and TNF- α (Figure 3E). By contrast, both p50 and c-Rel had already been recruited to the

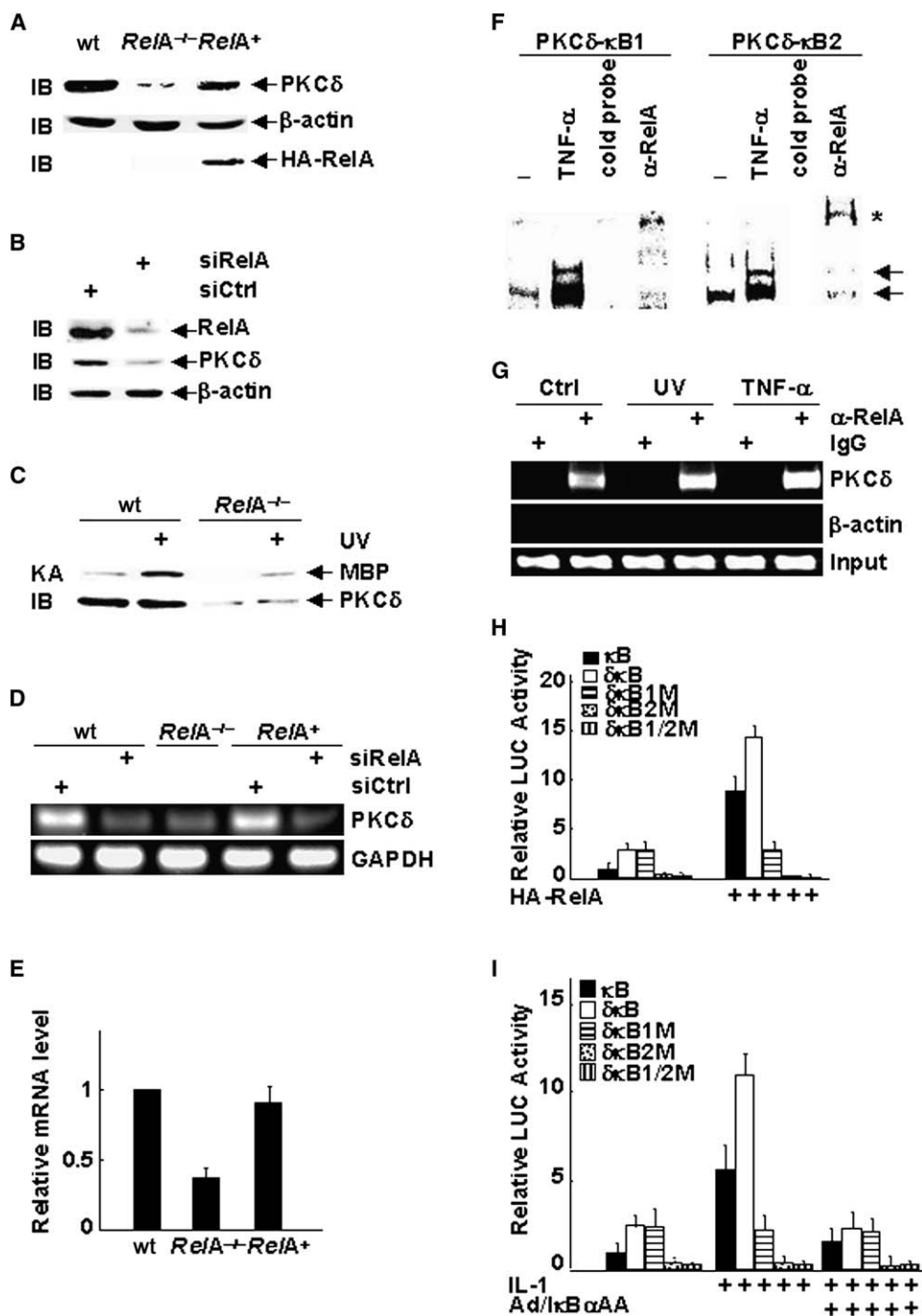


Figure 2. RelA Activity Is Required for PKC δ Expression

(A) Wt or *RelA*^{-/-} cells were transfected with expression vector encoding HA-RelA or empty vector (0.1 μ g each). Expressions of PKC δ and HA-RelA proteins were determined.

(B) Wt fibroblasts were transfected with the control or RelA siRNA (200 nM each). Expression levels of RelA and PKC δ proteins were determined.

(C) Wt and *RelA*^{-/-} cells were irradiated with or without UV (20 J/m², 30 min). PKC δ activity and expression were determined.

(D and E) Wt or *RelA*^{-/-} cells were transfected with expression vector encoding HA-RelA or empty vector (0.1 μ g each) along with or without the control or RelA siRNA (200 nM each). mRNA levels of PKC δ were measured by semiquantitative RT-PCR (D) or quantitative real-time RT-PCR (E). The results are presented as means $\pm$ standard errors and represent three independent experiments.

(F) Wt fibroblasts were treated with or without TNF- α (5 ng/ml) for 30 min. Whole-cell extracts were used for EMSA with ³²P-labeled PKC δ - κ B1- or PKC δ - κ B2-containing probes in the presence or absence of 100-fold excess of unlabeled probes or with or without anti-RelA antibody. Specific protein-DNA complexes were indicated by arrows, and the supershifted RelA-DNA complex was indicated by an asterisk.

(G) Wt fibroblasts were treated with or without UV (20 J/m², 4 hr) or TNF- α (5 ng/ml, 30 min). The recruitment of RelA to PKC δ promoter was examined by ChIP assays using anti-RelA antibody or IgG. β -actin was used as control. Input, 0.04% of total chromatin in immunoprecipitation.

(H and I) Wt fibroblasts were transfected with wt PKC δ -LUC reporter gene or its mutants (PKC δ - κ B1M-LUC, PKC δ - κ B2M-LUC, PKC δ - κ B1/2M-LUC), in which PKC δ - κ B1 and PKC δ - κ B2 sites have been mutated, individually or collectively, along with expression vector encoding HA-RelA or

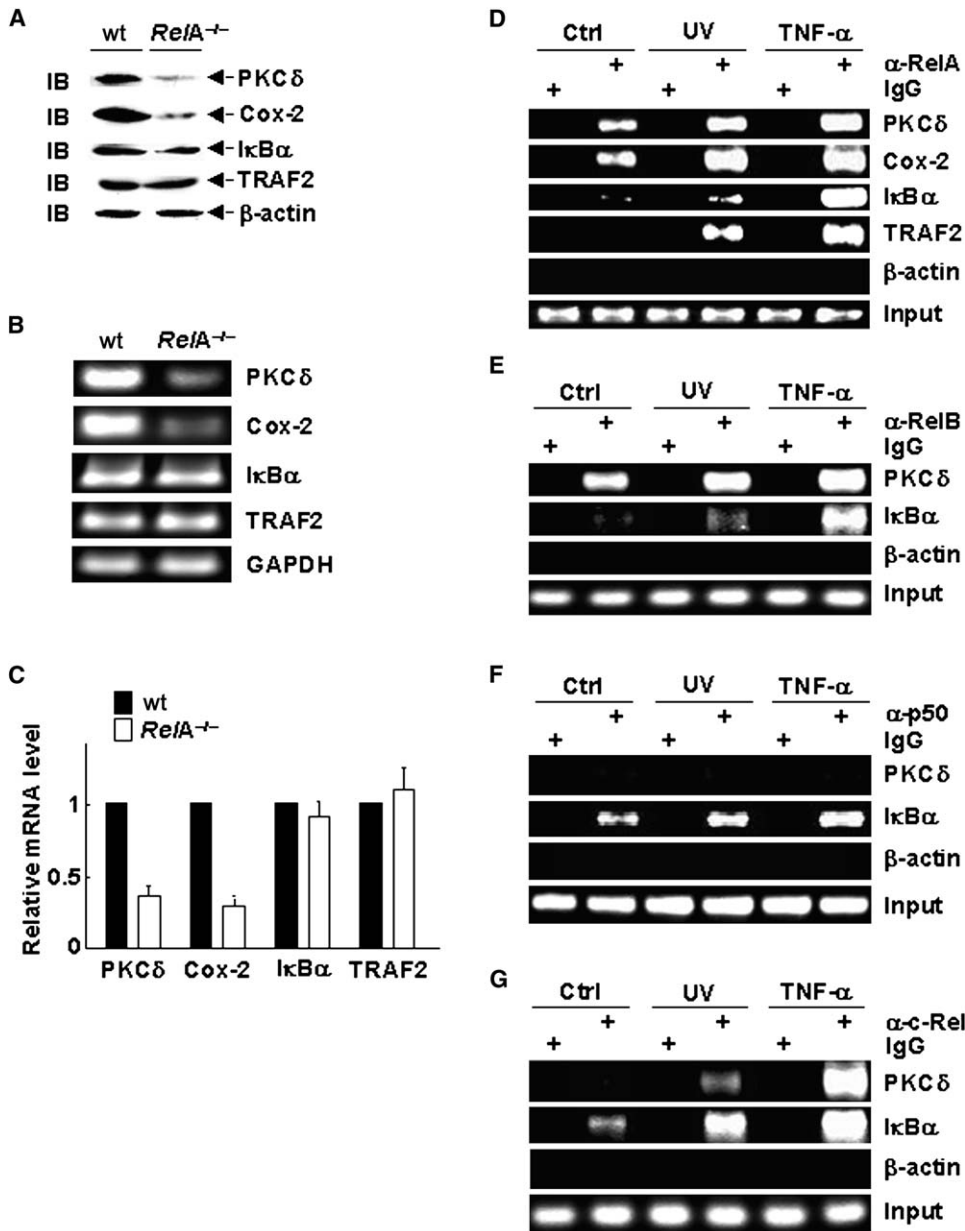


Figure 3. RelA Is Differentially Recruited to the Promoters of NF- κ B Target Genes

(A–C) Expression levels of PKCδ, Cox-2, IκBα, and TRAF2 proteins or mRNA in nonstimulated wt and RelA^{-/-} cells were analyzed by immunoblotting (A), semiquantitative RT-PCR (B), and quantitative real-time RT-PCR (C). The results are presented as means ± standard errors and represent three independent experiments.

(D–G) Wt fibroblasts were treated with or without UV (20 J/m², 4 hr) or TNF-α (5 ng/ml, 30 min). The recruitment of RelA to the promoters of PKCδ, Cox-2, IκBα, and TRAF2 (D) and the recruitment of RelB (E), p50 (F), or c-Rel (G) to the promoters of PKCδ and IκBα were examined by ChIP assays as described in Figure 2G.

promoter of IκBα, but not PKCδ, under nonstimulated conditions (Figures 3F and 3G). The recruitment of p50 to IκBα promoter, as well as the recruitment of c-Rel to both IκBα and PKCδ promoters, was further enhanced by UV and TNF-α (Figures 3F and 3G). Taken together, selective recruitment of individual NF- κ B subunits to the promoters of NF- κ B-target genes under nonstimu-

lated conditions appeared to determine whether their expressions were affected by loss of RelA.

PKCδ Is Required for JNK Activation by UV

To determine whether the NF- κ B target gene PKCδ is critical for UV-induced JNK activation, we used wt and Pkcδ^{-/-} fibroblasts (Figure 4A). Immune complex kinase

empty vector (0.1 μg each) (H), or treated with IL-1β (2 ng/ml, 10 hr) in the presence of Ad/IκBα(AA) or control Ad/LacZ (200 MOI each) (I). As control, 2 × NF- κ B-Luc reporter gene was used. LUC activity was determined (Liu et al., 2004). The results are presented as means ± standard errors and represent three independent experiments.

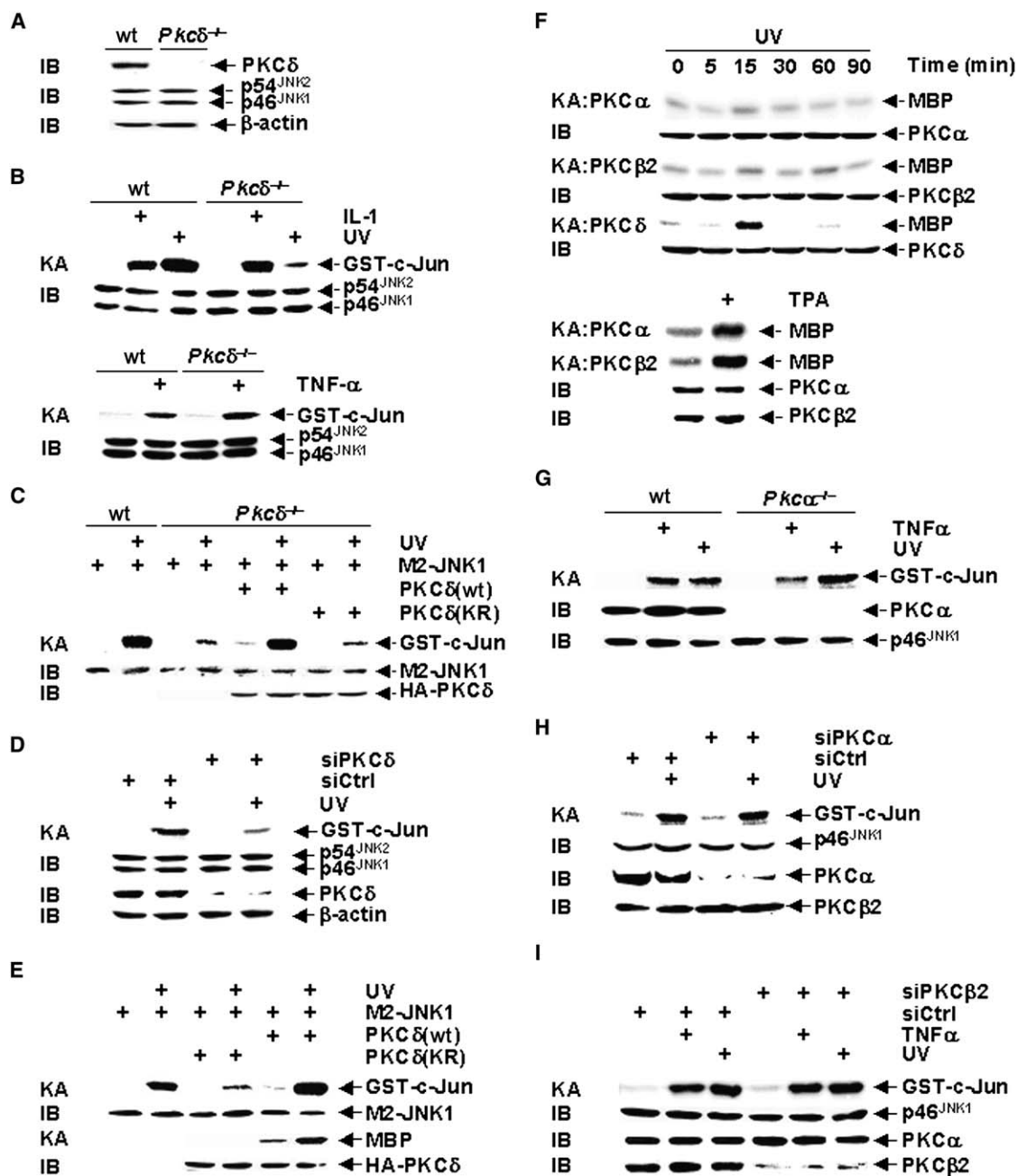


Figure 4. PKC δ Is Required for JNK Activation by UV

(A and B) Expression levels of PKC δ (A) and activation of JNK by UV (20 J/m², 30 min), IL-1 β (2 ng/ml, 15 min), or TNF- α (5 ng/ml, 15 min) (B) in wt and *Pkc* δ ^{-/-} cells were determined.

(C) *Pkc* δ ^{-/-} cells were transfected with expression vectors encoding M2-JNK1 (1 μ g) along with HA-PKC δ (WT), the kinase-deficient HA-PKC δ (KR) mutant or empty vector (2 μ g each). After 40 hr, cells were irradiated with or without UV (20 J/m², 30 min). M2-JNK1 activity and expression of M2-JNK1 and HA-PKC δ were determined. Wt fibroblasts were used as control.

(D) Wt fibroblasts were transfected with the control or PKC δ siRNA (200 nM each). UV-induced JNK activation and expression of JNK and PKC δ were determined as in (B) and (A).

(E) The effect of PKC δ on UV-induced JNK activation in wt fibroblasts was determined as in (C).

(F) Wt fibroblasts were irradiated with UV (20 J/m²) for various times as indicated or treated with TPA (50 ng/ml, 30 min). Activation and expression of PKC α , PKC β 2, or PKC δ were determined.

(G) Wt and *Pkc* α ^{-/-} cells were treated with UV (20 J/m², 30 min) or TNF α (5 ng/ml, 15 min). JNK activation and expression of JNK and PKC α were determined.

(H and I) Wt fibroblasts were transfected with the control, PKC α siRNA (H), or PKC β 2 siRNA (I) (200 nM each). UV-induced JNK activation and expression of JNK, PKC α , and PKC β 2 were determined.

assays showed that UV-induced JNK activation was severely impaired in *Pkc δ ^{-/-}* cells (Figure 4B). By contrast, JNK activation by IL-1 β or TNF- α was not affected by loss of PKC δ (Figure 4B). Ectopic expression of wt PKC δ but not its kinase-deficient PKC δ (KR) mutant restored UV-induced activation of cotransfected M2-JNK1 in *Pkc δ ^{-/-}* cells (Figure 4C). Similarly, UV-induced JNK activation in wt fibroblasts was suppressed by PKC δ siRNA, which inhibited PKC δ expression, but not the control scramble siRNA (Figure 4D). Conversely, UV-induced M2-JNK1 activation was synergized by cotransfected wt PKC δ , which itself was activated by UV, but inhibited by PKC δ (KR) mutant (Figure 4E). Loss of PKC δ did not affect UV-induced p38 activation (data not shown), consistent with the finding that UV-induced p38 activation was not affected by loss of RelA (Figure 1I). Thus, the NF- κ B target gene PKC δ is required for JNK activation by UV but not other stimuli like IL-1 β and TNF- α .

Next we tested whether PKC α and PKC β 2, which have been reported to facilitate JNK2 activation by UV, TPA, and TNF- α (Lopez-Bergami et al., 2005), are required for UV-induced JNK activation. Immune complex kinase assays showed that UV significantly activated PKC δ but only marginally activated PKC α or PKC β 2 (Figure 4F). Both PKC α and PKC β 2 could be significantly activated by TPA (Figure 4F). Consistently, UV-induced JNK activation was not impaired in *Pkc α ^{-/-}* cells (Figure 4G), whereas TNF- α -induced JNK activation was reduced in *Pkc α ^{-/-}* cells (Figure 4G). UV-induced JNK activation was also not suppressed by PKC α siRNA (Figure 4H) or PKC β 2 siRNA (Figure 4I). Thus, PKC δ but not PKC α or PKC β 2 is involved in UV-induced JNK activation.

PKC δ Is Essential for RelA to Positively Regulate UV-Induced JNK Activation

Although PKC δ is required for UV-induced JNK activation and its expression depends on RelA/NF- κ B, these findings do not mean that PKC δ is essential in mediating the augmentation of UV-induced JNK activation by RelA. To address this issue, we determined the effect of RelA on UV-induced JNK activation in *Pkc δ ^{-/-}* cells, in which UV-induced JNK activation was still detectable despite significant reduction (Figure 4B). Indeed, inhibition of RelA expression by RelA siRNA significantly suppressed UV-induced JNK activation in wt fibroblasts but was unable to do so in *Pkc δ ^{-/-}* cells (Figure 5A). Conversely, expression of HA-RelA promoted UV-induced activation of cotransfected M2-JNK1 in wt but not in *Pkc δ ^{-/-}* cells (Figure 5B). Similar results were obtained in PKC δ siRNA-expressing wt fibroblasts (Figure 5C). Expression of HA-PKC δ but not HA-PKC α in *RelA^{-/-}* cells restored activation of cotransfected M2-JNK1 by UV (Figure 5D). Taken together, PKC δ is necessary and sufficient to mediate the augmentation of UV-induced JNK activation by RelA.

Next, we determined how PKC δ regulates UV-induced JNK activation. Immunoblotting analysis revealed that UV-induced phosphorylation of JNK at Thr183 and Tyr185, which are required for JNK activation (Lin et al., 1995; Lu et al., 1997), was significantly reduced in *Pkc δ ^{-/-}* cells or PKC δ siRNA-expressing wt fibroblasts (Figures 6A and 6B). However, UV-induced phosphorylation of JNK1 and JNK2, the MAP2Ks of JNK, was not affected by loss of PKC δ (Figure 6C), indicating

that PKC δ may target JNKs or JNK but not MAP3Ks. Immune complex kinase assays showed that UV-activated PKC δ significantly phosphorylated GST-JNK1 but not GST-JNK1(S129A) mutant (Figure 6D), suggesting that PKC δ promoted JNK1 activation via phosphorylation at Ser129, a mechanism similar to the regulation of JNK2 by PKC α and PKC β 2 (Lopez-Bergami et al., 2005). Indeed, although the activity of both wt GST-JNK1 and GST-JNK1(S129A) mutant was stimulated by extracts of UV-irradiated *Pkc δ ^{-/-}* cells, only wt GST-JNK1 activity was further stimulated by UV-activated PKC δ , as analyzed by an in vitro reconstitution assay (Figure 6E). Under the same conditions, the kinase-deficient JNK1(APF) mutant, in which Thr183 and Tyr185 have been replaced by alanine and phenylalanine, respectively, and therefore can no longer be activated (Zheng et al., 1999), was not activated (Figure 6E). However, activated PKC δ itself did not stimulate GST-JNK1 activity (Figure 6E), suggesting that PKC δ only potentiated UV-induced JNK activation. This was also the case in living cells. Immune complex kinase assays and immunoblotting analysis revealed that PKC δ siRNA significantly inhibited UV-induced phosphorylation and activation of WT-JNK1 but only marginally affected the phosphorylation and activation of JNK1(S129A) mutant (Figure 6F). Taken together, the augmentation of UV-induced JNK activation by PKC δ is likely due, at least in part, to its phosphorylation of JNK1.

Positive Regulation of JNK Activity by RelA via PKC δ Is Critical for UV-Induced Apoptosis

The augmentation of UV-induced JNK activation by RelA/NF- κ B via PKC δ promoted us to examine whether the crosstalk affects UV-induced apoptosis. To test this idea, wt and *RelA^{-/-}* null cells were treated with UV irradiation or TNF- α or were left untreated. UV-induced apoptosis was significantly suppressed in *RelA^{-/-}* cells, as monitored by apoptotic death assays (Figure 7A). By contrast, loss of RelA promoted TNF- α -induced apoptosis (Figure 7A), as previously reported (Tang et al., 2001a). Similar results were obtained with Caspase-3 activity assays (see Figure S1 in the Supplemental Data available with this article online). Ectopic expression of HA-RelA along with enhanced green fluorescence protein (EGFP) (at a ratio of 4:1) (Tang et al., 2001b) in *RelA^{-/-}* cells restored UV-induced but suppressed TNF- α -induced apoptosis in EGFP-positive cells (Figure S2). Thus, RelA/NF- κ B contributes to UV-induced apoptosis.

To determine whether the promotion of UV-induced apoptosis by RelA/NF- κ B is due, at least in part, to its positive regulation of JNK activity, wt and *RelA^{-/-}* fibroblasts were irradiated with different doses of UV. At 10 J/m², UV efficiently induced apoptosis and c-Jun phosphorylation in wt fibroblasts but not in *RelA^{-/-}* cells, as monitored by apoptotic death assays (Figure 7B) and immunoblotting analysis (Figure S3). At 60 J/m², UV was able to induce apoptosis in *RelA^{-/-}* cells to the degree similar to that induced by 10 J/m² of UV in wt fibroblasts (Figure 7B). Immune complex kinase assays showed that JNK activity induced by 60 J/m² of UV in *RelA^{-/-}* cells was comparable to that induced by 10 J/m² of UV in wt fibroblasts (Figure 7C). These results suggest that the sensitivity of cells to UV-induced apoptosis depended

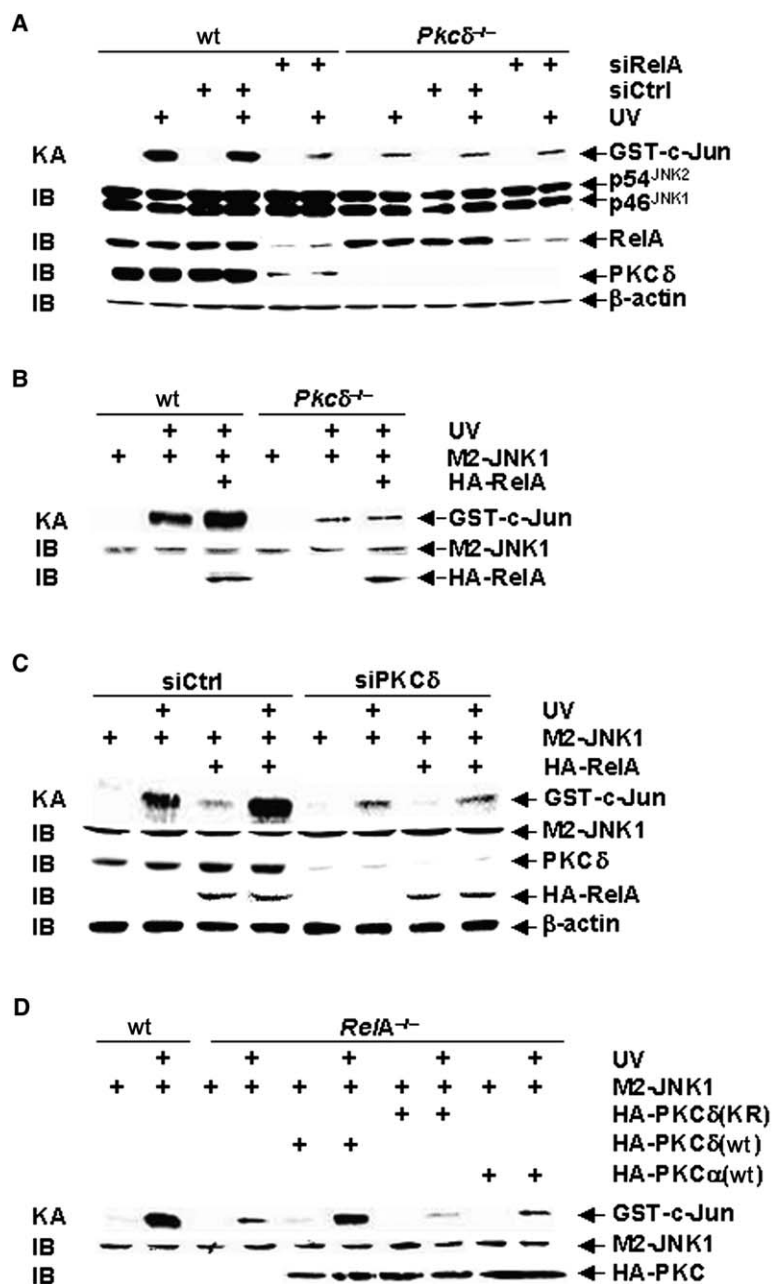


Figure 5. Augmentation of UV-Induced JNK Activation by RelA Is Mediated by PKCδ

(A and B) Wt and *Pkcδ*^{-/-} cells were transfected with the control or RelA siRNA (200 nM each) (A) or expression vectors encoding M2-JNK1 (1 μg), HA-RelA, or empty vector (0.1 μg each) (B). After 40 hr, cells were irradiated with or without UV (20 J/m², 30 min). JNK activity and expression of RelA, PKCδ, and JNK were determined.

(C) Wt fibroblasts were transfected with the control or PKCδ siRNA (200 nM each) along with M2-JNK1 (1 μg) and HA-RelA or empty vector (0.1 μg each). UV-induced M2-JNK1 activation and expression of PKCδ, M2-JNK1, and HA-RelA were determined as in (A) and (B).

(D) Wt or *RelA*^{-/-} cells were transfected with expression vectors encoding M2-JNK1 (1 μg) along with HA-PKCδ(WT), HA-PKCδ(KR) mutant, or HA-PKCα(WT) (2 μg each). UV-induced M2-JNK1 activation and expression of M2-JNK1 and HA-PKCs were determined.

on the strength of JNK activity. Consistently, transfection of the constitutively active JNKK2-JNK1 (Figure S4) (Zheng et al., 1999) along with EGFP (4:1 ratio) into *RelA*^{-/-} cells restored the sensitivity of EGFP-positive *RelA*^{-/-} cells to UV-induced apoptosis (Figure 7D). Similar results were obtained in *RelA*^{-/-} cells transfected with wt PKCδ, but not PKCδ(KR) mutant (Figure 7D). Thus, RelA/NF-κB determines the sensitivity of the cells to UV-induced apoptosis via PKCδ-mediated positive regulation of JNK activity.

Since PKCδ is essential for RelA to augment UV-induced JNK activation, we tested whether genetic disruption of PKCδ alleles suppresses UV-induced apoptosis and, if so, whether it is due to the reduction in JNK activation. Indeed, apoptotic death assays revealed that UV-induced apoptosis was impaired in *Pkcδ*^{-/-} cells

(Figure 7E). Expression of wt PKCδ, but not PKCδ(KR) mutant, along with EGFP (4:1 ratio) restored the sensitivity of EGFP-positive *Pkcδ*^{-/-} cells to UV-induced apoptosis (Figure S5). The proapoptotic effect of PKCδ in UV killing was mediated at least in part by JNK, as transfection of the constitutively active JNKK2-JNK1, but not JNKK2(KM)-JNK1 (Figure S6) (Zheng et al., 1999), along with EGFP (4:1 ratio) restored the sensitivity of EGFP-positive *Pkcδ*^{-/-} cells to UV killing (Figure 7F). Taken together, PKCδ is required for UV-induced apoptosis via positive regulation of JNK activity.

RelA-Augmented JNK Activation Is Required for Immediate Induction of Fas Ligand by UV

Activated NF-κB has been shown to contribute to UV-induced apoptosis via induction of the proapoptotic

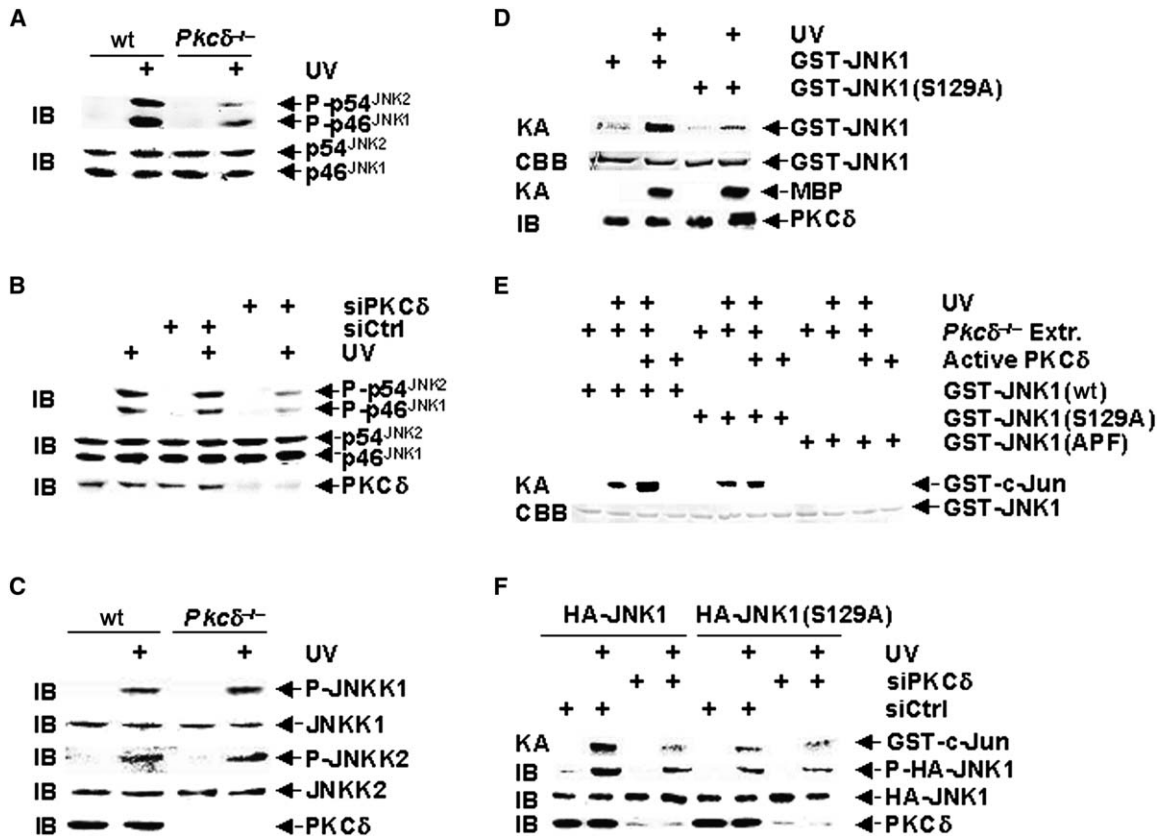


Figure 6. PKC δ Is Required for UV-Induced JNK Activation through Phosphorylation of JNK

(A) Wt and *Pkcδ*^{-/-} cells were irradiated with or without UV (20 J/m², 30 min). Phosphorylation and expression of JNK were determined. (B) Wt fibroblasts were transfected with the control or PKC δ siRNA (200 nM each). UV-induced phosphorylation of JNK and expression of JNK and PKC δ were determined. (C) Phosphorylation and expression of JNKK1, JNKK2, and PKC δ were determined in UV-irradiated wt and *Pkcδ*^{-/-} cells as in (A). (D) Phosphorylation of purified GST-JNK1, GST-JNK1 (S129A), or GST-JNK1(APF) mutant proteins by UV-activated or control PKC δ was determined by immune complex kinase assays. CBB, Coomassie brilliant blue. (E) Purified GST-JNK1, GST-JNK1 (S129A), or GST-JNK1(APF) mutant proteins were first incubated with or without UV-activated PKC δ in a kinase reaction containing nonradioactive ATP (100 μ M). GST-JNK1 proteins were bound to GSH beads, washed, and incubated with or without control or UV-irradiated *Pkcδ*^{-/-} cell extracts in a kinase reaction containing nonradioactive ATP (100 μ M). The modified GST-JNK1 proteins were used to phosphorylate GST-c-Jun as described in (Figure 1D). (F) Wt fibroblasts were transfected with expression vectors encoding HA-JNK1 or HA-JNK1 (S129A) along with the control or PKC δ siRNA (200 nM each). UV-induced activation and phosphorylation of HA-JNK1 and expression of HA-JNK1 and PKC δ were determined.

Fas ligand (FasL) (Kasibhatla et al., 1998) or suppression of the antiapoptotic Bcl-2 family protein Bcl-x_L expression (Campbell et al., 2004). Yet it has been puzzling how the slow NF- κ B activation by UV could possibly be involved in the rapid onset of the apoptosis. We tested whether the augmentation of UV-induced JNK activation by the preexisting nuclear RelA/NF- κ B is involved in the onset of UV-induced apoptosis. In wt fibroblasts, UV induced expression of FasL, which occurred as early as 1 hr (the early phase) and lasted at least 8 hr (the late phase), as analyzed by semiquantitative RT-PCR (Figure 7G). Interestingly, UV-induced FasL expression was diminished in *RelA*^{-/-} cells (Figure 7G). Introduction of PKC δ or the constitutively active JNKK2-JNK1 into *RelA*^{-/-} cells partially restored the early-phase but not the late-phase induction of FasL (Figure 7G). Consistently, UV-induced early phase induction of FasL was significantly delayed in *Pkcδ*^{-/-} cells (which had weakened JNK activity) or

Jnk1^{-/-} cells (which had little JNK activity) (Figure 7G). This suggests that the augmentation of UV-induced JNK activation by the preexisting nuclear RelA via PKC δ was essential for the early phase induction of FasL, whereas both activated RelA/NF- κ B and c-Jun/AP-1 were required for the late phase induction of FasL (Kasibhatla et al., 1998). Expression of Bcl-x_L was suppressed by UV irradiation in wt fibroblasts when compared to that in *RelA*^{-/-} cells, as analyzed by semiquantitative RT-PCR (Figure 7H), consistent with the previous report (Campbell et al., 2004). However, UV-induced suppression of Bcl-x_L expression was not affected by loss of PKC δ or JNK1 (Figure 7H), suggesting that the RelA/NF- κ B-PKC δ -JNK pathway may not be involved in suppression of Bcl-x_L expression. Taken together, the augmentation of UV-induced JNK activation by the preexisting nuclear RelA is required for immediate induction of FasL expression during UV-induced apoptosis.

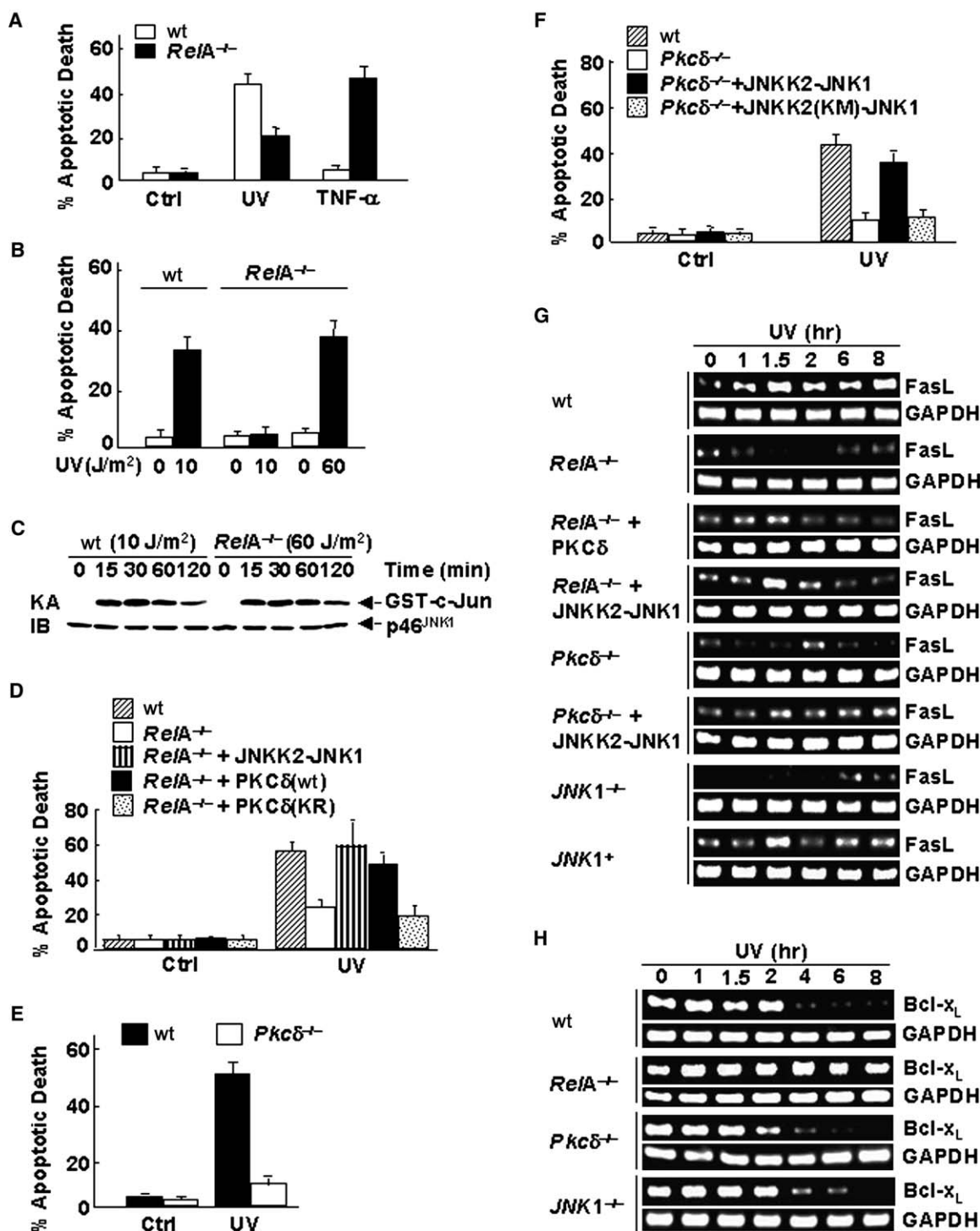


Figure 7. Augmentation of JNK Activation by the Preexisting Nuclear RelA/NF-κB Is Required for UV-Induced Apoptosis through Immediate Induction of Fas Ligand

(A) Wt and *RelA*^{-/-} cells were treated with or without UV (20 J/m²) or TNF-α (5 ng/ml) for 24 hr. Apoptotic cell death was measured with Hoechst 33258 staining.

(B and C) Wt and *RelA*^{-/-} cells were treated with or without UV at the doses of 10 J/m² or 60 J/m² for 24 hr and the apoptotic cell death determined (B), or for 30 min, and JNK activation and expression were determined (C).

(D) Wt or *RelA*^{-/-} cells were transfected with expression vectors encoding HA-PKCδ(WT), HA-PKCδ(KR), HA-JNKK2-JNK1, or empty vector (3.2 μg each), along with EGFP (0.8 μg). UV-induced apoptotic death of EGFP-positive cells was determined as in (A).

(E) UV-induced apoptotic cell death was determined in wt and *Pkcδ*^{-/-} cells as in (A).

(F) Wt or *Pkcδ*^{-/-} cells were transfected with expressing vectors encoding HA-JNKK2-JNK1, HA-JNKK2(KM)-JNK1, or empty vector (3.2 μg each) along with EGFP (0.8 μg). UV-induced apoptotic cell death of EGFP-positive cells was determined as in (A). The results of (A)–(F) are presented as means ± standard errors and represent three independent experiments.

Discussion

UV exerts its biological functions through activation of multiple downstream effectors, including NF- κ B, JNK, and caspases (Herrlich et al., 1997). UV induces rapid and robust activation of JNK1 (Hibi et al., 1993; Liu et al., 2004), which is essential for UV-induced apoptosis (Liu et al., 2004; Tournier et al., 2000). By contrast, UV activates NF- κ B in much slower kinetics via the p38-CK2 pathway (Bender et al., 1998; Li and Karin, 1998; Kato et al., 2003). It is unknown whether any crosstalk occurs between NF- κ B and JNK in response to UV and, if so, how it affects UV-induced apoptosis. In this report, we demonstrate that the preexisting nuclear RelA augments UV-induced JNK activation via induction of PKC δ expression, thereby contributing to UV-induced apoptosis. This conclusion is based on the following evidence.

First, genetic disruption of RelA alleles or siRNA-mediated inhibition of RelA expression significantly reduced UV-induced JNK activation in mouse fibroblasts (Figure 1). Second, in nonstimulated cells, the preexisting nuclear RelA had already selectively been recruited to PKC δ promoter and was essential for its expression (Figures 2 and 3). Third, PKC δ but not PKC α or PKC β 2 was essential for mediating the augmentation of UV-induced JNK activation by RelA through phosphorylation of JNK1 at Ser129 (Figures 4, 5, and 6). Fourth, the augmentation of UV-induced JNK activation by RelA was responsible for immediate FasL expression (Figure 7), which is involved in the initiation of UV-induced apoptosis. Taken together, our results provide a mechanism by which RelA via PKC δ augments UV-induced JNK activation for the onset of UV-induced apoptosis.

Our finding that UV-induced JNK activation depends on the preexisting nuclear RelA/NF- κ B is quite surprising. Although it has been shown that I κ B proteins constantly undergo signal-independent proteolysis (Pando and Verma, 2000) and NF- κ B does shuttle between the cytoplasm and nucleus (Ghosh and Karin, 2002), most of the studies on the biological functions of NF- κ B have been focused on the role of activated NF- κ B in various cellular activities. Our results show that the preexisting nuclear RelA/NF- κ B promoted UV-induced JNK activation and apoptosis (Figures 1 and 7) via induction of PKC δ (Figures 2 and 3), which was a key component of the UV-JNK signaling pathway and was essential for mediating the augmentation of UV-induced JNK activation by RelA (Figures 4, 5, and 6). By contrast, activated NF- κ B may not be critical for UV-induced JNK activation, as blockade of UV-induced I κ B α degradation and NF- κ B activation by the p38 inhibitor SB208590 did not affect UV-induced JNK activation (data not shown). Although activated NF- κ B further enhanced PKC δ expression (Figures 2 and 3), the preexisting nuclear RelA-induced expression of PKC δ is likely already sufficient for JNK activation by UV. Thus, the upregulation of PKC δ by the preexisting nuclear RelA/NF- κ B is crucial

in priming cells so that they have rapid and robust JNK activation in response to UV irradiation.

How can NF- κ B augment UV-induced JNK activation, given the fact that it inhibits TNF- α -induced JNK activation (Tang et al., 2001a)? One of the explanations is that, while activated NF- κ B was required for inhibition of TNF- α -induced JNK activation (Tang et al., 2001a), the preexisting nuclear RelA activity was responsible for the augmentation of UV-induced JNK activation (Figures 1–5). Activation of NF- κ B by TNF- α is rapid, which allows it to inhibit concomitantly activated JNK (Tang et al., 2001a; data not shown). By contrast, activation of NF- κ B by UV occurred 2–3 hr after the irradiation and thereby was unable to prevent UV-induced JNK activation, which reached maximum only 30 min after the irradiation (Figures 1B–1D). However, the preexisting nuclear RelA activity was sufficient to maintain PKC δ expression, which was essential for rapid and robust JNK activation by UV (Figures 2–5). Another explanation is that NF- κ B may regulate JNK activity in a signalling-pathway-dependent manner. The inhibition of JNK by NF- κ B is specific to TNF- α but not to other stimuli such as IL-1 β and UV (Tang et al., 2001a; Lin, 2003; data not shown). Interestingly, the augmentation of JNK activation by RelA via PKC δ was specific to UV but not to TNF- α or IL-1 β (Figures 1, 4, 5, and 6). Future studies will elucidate how PKC δ is differentially involved in JNK activation by different extracellular stimuli.

How is PKC δ expression regulated by RelA? Our results show that PKC δ was a bona fide RelA/NF- κ B target gene (Figure 2) and that its expression depended on the preexisting nuclear RelA/NF- κ B (Figures 2G and 3). In nonstimulated cells, RelA had already been recruited to the promoters of PKC δ as well as Cox-2 (Figures 2G and 3D), but not I κ B α and TRAF2, whose expression were not affected by loss of RelA (Figures 3A–3C). Like RelA, RelB had also been recruited to the promoter of PKC δ , but not I κ B α in nonstimulated cells (Figure 3E). By contrast, both p50 and c-Rel were recruited to the promoter of I κ B α , but not PKC δ (Figures 3F and 3G). These observations suggest that the affinities of individual NF- κ B subunits toward NF- κ B-dependent promoters are different. Differential recruitment of individual NF- κ B subunits to NF- κ B-dependent promoters under nonstimulated conditions may determine whether expressions of some NF- κ B target genes like PKC δ and Cox-2, but not others like I κ B α and TRAF2, were affected by loss of RelA. Upon stimulation, RelA was recruited to all four promoters (Figure 3D). It is likely that the increased amount or posttranslational modification(s) of nuclear RelA proteins upon stimulation may allow it to interact with the low-affinity promoters as well. Future studies will test these possibilities.

PKC δ is necessary and sufficient in mediating the augmentation of UV-induced JNK activation by RelA. Our results show that UV strongly activated PKC δ (Figure 4F) and that UV-induced JNK activation was impaired in *Pkc δ ^{-/-}* cells or in wt fibroblasts expressing

(G) Wt, *RelA*^{-/-}, *Pkc δ* ^{-/-}, and *JNK1*^{-/-} cells were transfected with or without PKC δ , HA-JNKK2-JNK1, HA-JNK1, or empty vector, followed by UV irradiation (20 J/m²) for various times as indicated. mRNA levels of FasL were analyzed by semiquantitative RT-PCR. GAPDH was used as control.

(H) Wt, *RelA*^{-/-}, *Pkc δ* ^{-/-}, and *JNK1*^{-/-} cells were irradiated with or without UV (20 J/m²) as in (G). mRNA levels of Bcl-x_L were analyzed by semiquantitative RT-PCR. GAPDH was used as control.

PKC δ siRNA (Figures 4B–4D). The augmentation of UV-induced JNK activation by RelA required PKC δ (Figures 5A–5C), which promoted JNK phosphorylation-activation through, at least in part, direct phosphorylation of JNK1 at Ser129 (Figures 6D–6F). By contrast, PKC α and PKC β 2, which are reported to facilitate JNK2 activation by UV, TPA, and TNF- α via Ser129 phosphorylation (Lopez-Bergami et al., 2005), were only marginally activated by UV (Figure 4F). More importantly, UV-induced JNK activation was not inhibited by loss of PKC α or PKC β 2 in murine fibroblasts (Figures 4G–4I). Future studies will elucidate how PKC isoforms augment JNK activation in different cellular contexts and in response to different stimuli.

The augmentation of JNK activation by RelA is critical for UV-induced apoptosis. In contrast to its antiapoptotic functions in response to most death stimuli, NF- κ B has been reported to promote UV-induced apoptosis (Kasibhatla et al., 1998; Campbell et al., 2004). However, it has been puzzling how slow activation of NF- κ B by UV (Figures 1B and 1C; Bender et al., 1998; Li and Karin, 1998) could possibly contribute to the rapid initiation of UV-induced apoptosis (Nijhawan et al., 2003). Our results show that the preexisting nuclear RelA/NF- κ B played a critical role in UV-induced rapid activation of JNK, which is essential for the initiation of UV-induced apoptosis (Liu et al., 2004; Tournier et al., 2000). UV-induced apoptosis was significantly reduced in cells deficient in RelA (Figure 7A) or PKC δ (Figure 7E), in which JNK activation was significantly reduced (Figures 1D and 4B). However, *RelA*^{-/-} cells had death rates similar to wt fibroblasts when both cells had a similar degree of JNK activation (Figures 7B and 7C). Thus, RelA/NF- κ B promotes UV-induced apoptosis via regulation of the strength of JNK activation (Figure 7).

How does the augmentation of UV-induced JNK activation by RelA contribute to UV killing? Green and his colleagues have shown that, in Jurkat T cells, activated NF- κ B in combination with c-Jun/AP-1 promotes UV killing through induction of FasL (Kasibhatla et al., 1998). Activated RelA/NF- κ B has also been reported to suppress Bcl-x_L expression (Campbell et al., 2004). Our results show that UV-induced expressions of FasL, both early and late phases, were abolished in *RelA*^{-/-} cells (Figure 7G). Introduction of PKC δ or the active JNK1 (JNKK2-JNK1), both of which restored JNK activation (Figure 5D and Figure S4), into *RelA*^{-/-} cells restored only the early phase of UV-induced FasL expression (Figure 7G). Thus, the early phase of UV-induced FasL expression depended on the augmentation of JNK activation by RelA. In support of this notion, the early phase of UV-induced FasL expression was significantly delayed in cells deficient in PKC δ or JNK1 (Figures 7G), in which NF- κ B activation was not affected (data not shown; Figure 7H). By contrast, the suppression of Bcl-x_L expression by UV, which depends on activated NF- κ B (Campbell et al., 2004), was not affected in cells deficient in PKC δ or JNK1 (Figure 7H). Thus, the promotion of UV-induced apoptosis by NF- κ B is at least 2-fold. First, the preexisting nuclear RelA/NF- κ B via PKC δ augments JNK activation by UV, which is required for the early phase of UV-induced FasL expression (Figures 7G). The activation of JNK is also critical for the proapoptotic activity of p53 (Shaulian et al., 2000). Second,

activated NF- κ B suppresses expression of Bcl-x_L (Figure 7H; Campbell et al., 2004) and meanwhile further participates in induction of FasL expression in cooperation with c-Jun/AP-1 (Kasibhatla et al., 1998), thereby reinforcing the apoptotic process.

Conclusion Remarks

Previously, it has been shown that, in addition to inhibition of caspases, activated NF- κ B inhibits TNF- α -induced JNK activation, thereby suppressing TNF- α killing (Karin and Lin, 2002; Lin, 2003; Liu et al., 2004; Tang et al., 2001a). Here we show that the preexisting nuclear RelA/NF- κ B via PKC δ augments UV-induced JNK activation, thereby contributing to UV-induced apoptosis. Thus, depending on the nature of death stimuli, NF- κ B can either positively regulate JNK activity for apoptosis or negatively regulate JNK activity for cell survival. Future studies on how NF- κ B regulates JNK activity in response to various extracellular stimuli should provide a better understanding of the JNK biology.

Experimental Procedures

Reagents

Antibodies against JNK (333 for immunoprecipitation and 666 for immunoblotting) were from Pharmingen. Antibodies against phospho-JNK, p38, phospho-p38, JNKK1, JNKK2, phospho-JNKK1, phospho-JNKK2, and phospho-c-Jun (Ser63) were from Cell Signaling. Antibody against RelA was from Zymed. Antibodies against RelA (sc-372X), RelB (sc-226X), p50 (sc-7178X), c-Rel (sc-272X), I κ B α , TRAF2, Cox-2, PKC δ , PKC α , PKC β 2, and hemagglutinin (HA) were from Santa Cruz. Nocodazole, Hoechst 33258, myelin basic protein (MBP), TPA, and antibodies against M2-tag and β -actin were from Sigma. IL-1 β and murine TNF- α were from R&D Systems. Caspase-3 substrate VII (DEVD-AFC) was from Calbiochem. [γ -³²P]ATP (3000 mCi/nmol) was from Dupont NEN.

Plasmids

Expression vectors encoding GST-JNK1, GST-JNK1(APF), M2-JNK1, HA-JNKK2-JNK1, HA-JNKK2(KM)-JNK1, EGFP, HA-RelA, HA-PKC δ (WT), HA-PKC δ (KR), HA-PKC α (WT), and adenoviral vectors encoding the superrepressor I κ B α (AA) or LacZ have been described previously (Corbit et al., 2003; Tang et al., 2001a; Tang et al., 2002; Liu et al., 2004). S129A mutation was introduced into JNK1 using the QuikChange Site-Directed Mutagenesis Kit (Stratagene). pGL2-Enhancer vector was from Promega. The PKC δ promoter region (-309 to -1) was subcloned between SacI and BglII sites of pGL2-Enhancer vector (PKC δ -LUC). κ B sites in the PKC δ promoter region were mutated individually or collectively (κ B1 site, -83 to -74, GGGAAGCCCC to TTGAAGCCCC; κ B2 site, -52 to -43, GGGGAGTCCC to GTTGAGTCCC) and subcloned between SacI and BglII sites of pGL2-Enhancer vector (named as PKC δ - κ B1M-LUC, PKC δ - κ B2M-LUC, and PKC δ - κ B1/2M-LUC). All constructs were verified by DNA sequencing.

Cell Culture, Transfection, and Reporter Gene Assays

Wt, *RelA*^{-/-}, *Pkc δ* ^{-/-}, or *Pkc α* ^{-/-} mouse fibroblasts were described previously (Tang et al., 2001a; Leitges et al., 2001, 2002). Cell culture conditions, transfections, and luciferase reporter gene assays have been described previously (Liu et al., 2004). SiRNAs were transfected using Lipofectamine 2000.

RNA Extraction, Semiquantitative RT-PCR, and Real-Time PCR

Total RNA was used for semiquantitative and quantitative real-time RT-PCR. GAPDH was used as an internal control. The primers were designed as the following: PKC δ -forward, CAGACCAAGGAC CACCTGTT; PKC δ -reverse, GCATAAAACGTAGCCCCGTA; Cox-2-forward, AGAAGGAAATGGCTGCAGAA; Cox-2-reverse, GCTCGGC TTCCAGTATTGAG; I κ B α -forward, AGGACGAGGAGTACGAGCAA; I κ B α -reverse, GTCTCCCTTCACCTGACCAA; TRAF2-forward, GCC AGATTTTGAGAGAGAAG; TRAF2-reverse, CCAGCTGTTGCACCT

TGTTA; FasL-forward, CATCACAACTCCACTG; FasL-reverse, GTTCTGCCAGTTCTCTG; Bcl-x_L-forward, TTCGGGATGAGTAACTGG; Bcl-x_L-reverse, TGCAATCCGACTACCAATA; GAPDH-forward, AACTTTGGCATTGTGAAGG; and GAPDH-reverse, GGATGCAGGGATGATGTTCT.

Electrophoretic Mobility Shift Assays

Electrophoretic mobility shift assays were performed with whole-cell extracts. The double-stranded oligonucleotides contained 5'-AGCT overhangs. The sense sequences used were the following: PKC δ - κ B1, CAGTGGGGAGTCCCGGG; PKC δ - κ B1M, CAGTGTTGAGTCCCGGG; PKC δ - κ B2, GTTGGGAAGCCCCGTC; and PKC δ - κ B2M, GTTGTGAAGCCCCGTC. The NF- κ B consensus sequences are underlined. For supershift analysis, anti-RelA antibody (2 μ g) was added prior to the addition of radiolabeled oligonucleotides. Samples were separated on 5% nondenaturing gels and analyzed by autoradiography (Lin et al., 1992).

Chromatin Immunoprecipitation Assays

Cells were stimulated with or without murine TNF- α (5 ng/ml, 30 min) or UV (20 J/m², 4 hr) and crosslinked with 1% formaldehyde at 37°C for 10 min. Cell extracts were prepared and sonicated to obtain DNA fragments with sizes between 0.2 and 0.5 kb. Protein-DNA complexes were immunoprecipitated using anti-RelA antibody (5 μ g, sc-372x; Santa Cruz) or IgG control. DNAs were purified and used for PCR with primers specific for the gene promoter region encompassing NF- κ B binding sites. The following primers were used: PKC δ , ACCTCCAGCTCCTTCTCTC and TGGGTCCCAGATCTGTAG; Cox-2, CAGATGTGGACCTGACAGA and ACGCAAATGAGAGACGAAGG; I κ B α , GGACCCCAACCAAAATCG and TCAGGC GCGGGGAATTTC; TRAF2, GCCCTAGTGTGGGAACAAA and CTGCTCCATCAAGACCCACT; and β -actin, TGCAGTGTGCGGC GAAGC and TCGAGCCATAAAAGGCAA.

Design of siRNA Specific for Murine RelA and Pkc δ

siRNAs specific for murine RelA and Pkc δ were designed and confirmed for uniqueness using the BLASTN program. The target sequence for RelA is 5'-GAGTTTCAGCAGCTCCTGAAC-3', for Pkc δ is 5'-AACTCACCAGATTCAAGGTTTA-3', for Pkc α is 5'-GAACGTG CATGAGGTGAAA-3', and for Pkc β 2 is 5'-GGATGAACTGACC GATTT.

Protein Kinase Assays, Immunoblotting, and Immunofluorescence

Immune complex kinase assays and immunoblotting were performed as described (Liu et al., 2004). Immunofluorescence was performed as described (Tang et al., 2001b).

Apoptosis Assays

Apoptotic death assays and Caspase-3 assays were described previously (Liu et al., 2004).

Supplemental Data

Supplemental Data include six figures and can be found with this article online at <http://www.molecule.org/cgi/content/21/4/467/DC1/>.

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